



Documentation

Foreign Language Learning Platform

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Application Description

EasyLingo is a desktop application designed to support foreign language learning, developed using WPF (Windows Presentation Foundation). The application allows users to learn vocabulary through word sets, multiple revision modes, and a system of progress tracking and achievements.

The application supports user registration and login, which enables multiple users to use the program on a single computer while maintaining separate learning progress. User data and vocabulary sets are stored locally in a SQLite database.

The goal of the project was to create a modern and intuitive educational application that combines learning with motivational elements such as progress percentages, completed sets, and achievements.

Functionalities

- **User Registration and Login** – allows users to create accounts and log in with data validation. This enables multiple users to learn different languages on the same computer and track individual progress.
- **Language Management** – the ability to select a learning language, dynamically switch languages in the application, and display only sets and categories assigned to the selected language.
- **Vocabulary Sets** – creating custom vocabulary sets, assigning sets to categories, editing and deleting sets, filtering sets by category and language, and importing/exporting sets to JSON files.
- **Term Management** – adding terms to a set, editing and deleting existing terms, and automatic recalculation of progress after each change.
- **Learning Modes** – for each set, four revision modes are available: flashcards, multiple-choice questions, matching words with translations, and manual typing of translations.
- **Progress and Achievements** – percentage progress for each set, overall user progress, number of completed sets, and an achievement and milestone system (e.g. first completed set).

Application Architecture

The application follows the MVVM (Model–View–ViewModel) architecture pattern:

- **View (XAML)** – responsible for the user interface,
- **ViewModel** – contains application logic and commands,
- **Model/Data** – entities, database, and data access.

Data access logic is separated into services and repositories, which improves readability and scalability of the application.

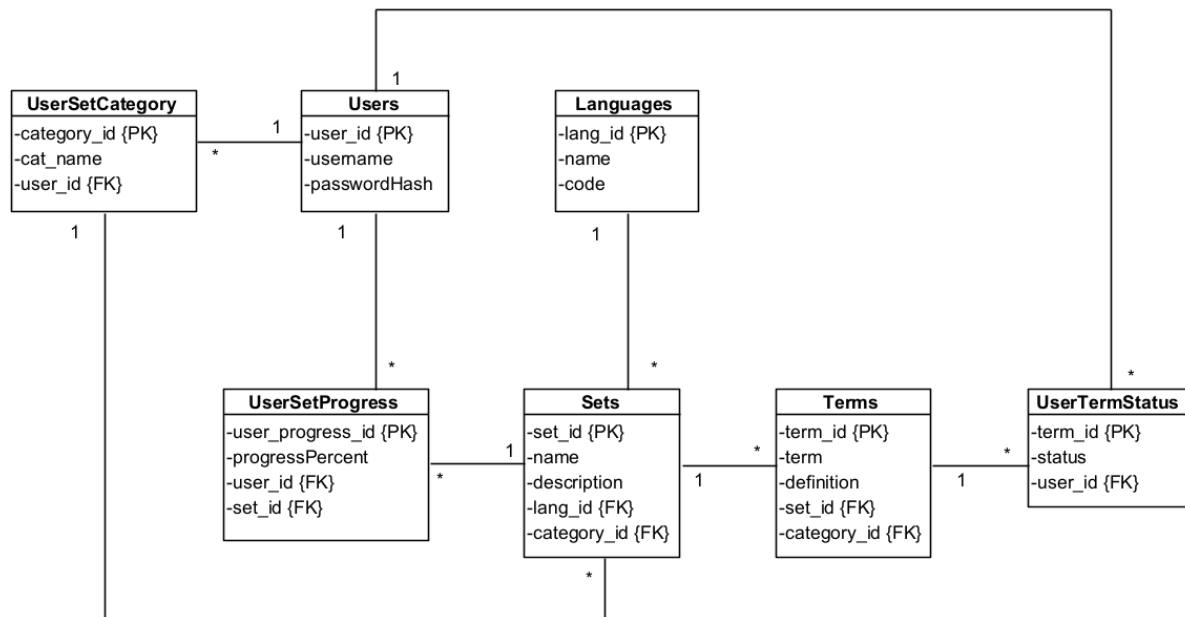
Technologies Used

- **.NET** – runtime environment and application framework on which the entire application is based. It provides support for WPF, asynchronous programming, LINQ, and database integration.
- **C#** – the primary programming language used to implement application logic.
- **WPF (Windows Presentation Foundation)** – a technology for building desktop application user interfaces, enabling separation of UI and logic (MVVM).
- **Entity Framework Core** – ORM tool used for communication with the SQLite database.
- **SQLite** – a lightweight local database used to store users, sets, terms, and progress data.

Database

Data is stored locally in a SQLite database. The data model includes:

- Users
- Vocabulary sets
- Terms
- Set categories
- User progress
- Term knowledge statuses
- Languages



1 Database design

Repositories and Services

The project uses generic repositories:

- IRepository<T> – repository interface,
- EfRepository<T> – implementation based on Entity Framework Core.

Application logic is separated into services:

- DataService – main data access and business logic,
- SetJsonPortService – responsible for importing and exporting sets to JSON files.

This approach improves code readability and follows object-oriented programming principles.

JSON Set Import and Export

The application supports saving and loading vocabulary sets using JSON files. This functionality allows:

- Creating backups,
- Transferring sets between users,
- Importing ready-made sets.

JSON File Structure

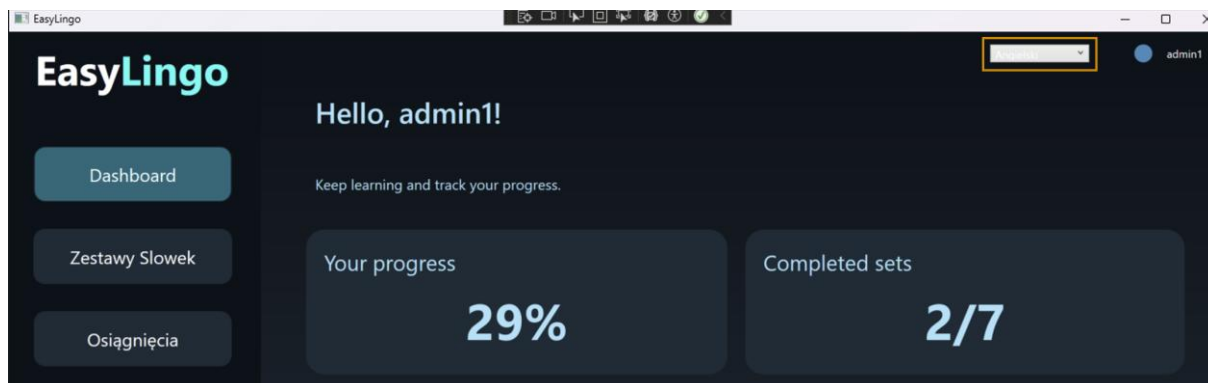
```
{
  "Version": 1,
  "Name": "Animals",
  "Description": "Basic animal vocabulary",
  "LanguageCode": "EN",
  "CategoryName": "Basics",
  "Terms": [
    {
      "TermName": "Dog",
      "Definition": "Pies"
    },
    {
      "TermName": "Cat",
      "Definition": "Kot"
    }
  ]
}
```

JSON Field Description:

- **Version** – file format version (future-proofing in case new fields are added, e.g. difficulty level or creation date),
- **Name** – set name (required),
- **Description** – set description,
- **LanguageCode** – language code,
- **CategoryName** – category name,
- **Terms** – list of vocabulary terms.

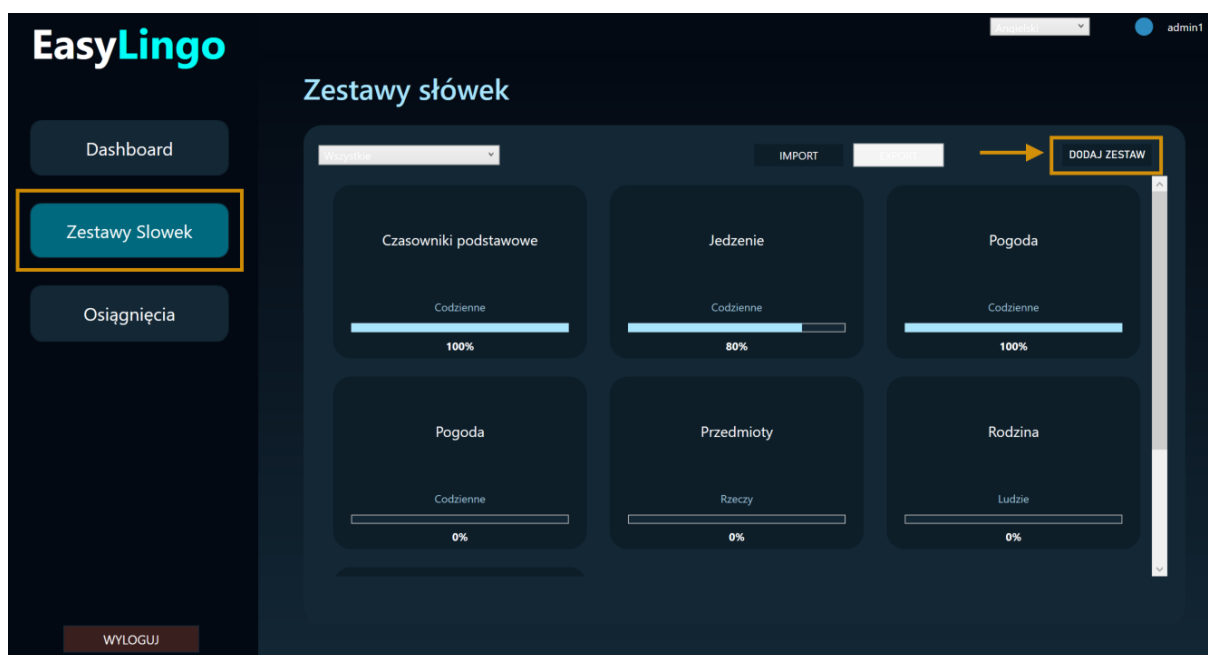
User Manual

1. Language Change

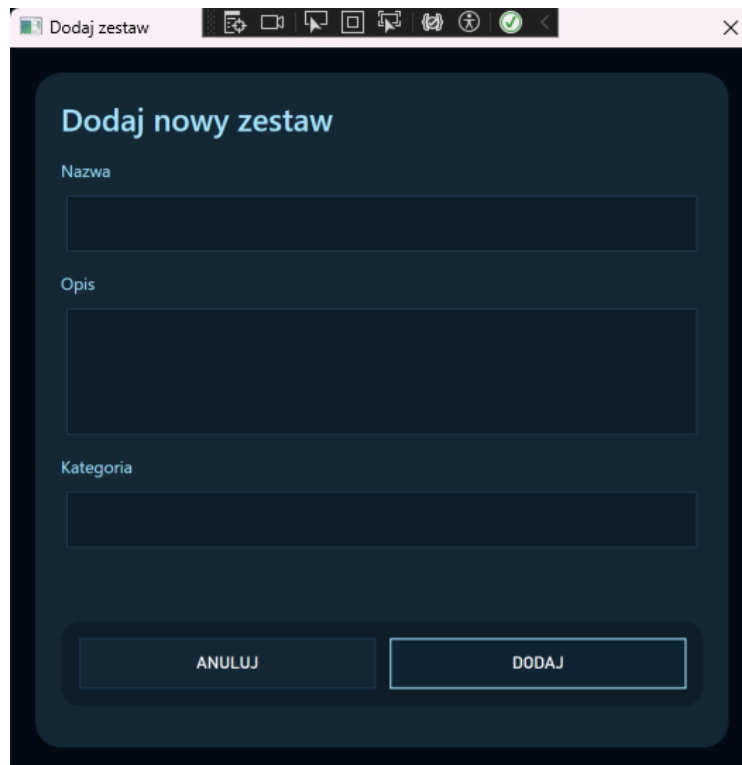


2. Adding a Set

Zestawy Słówek > Dodaj zestaw



The user must fill in the form and click the “DODAJ” button



Dodaj nowy zestaw

Nazwa

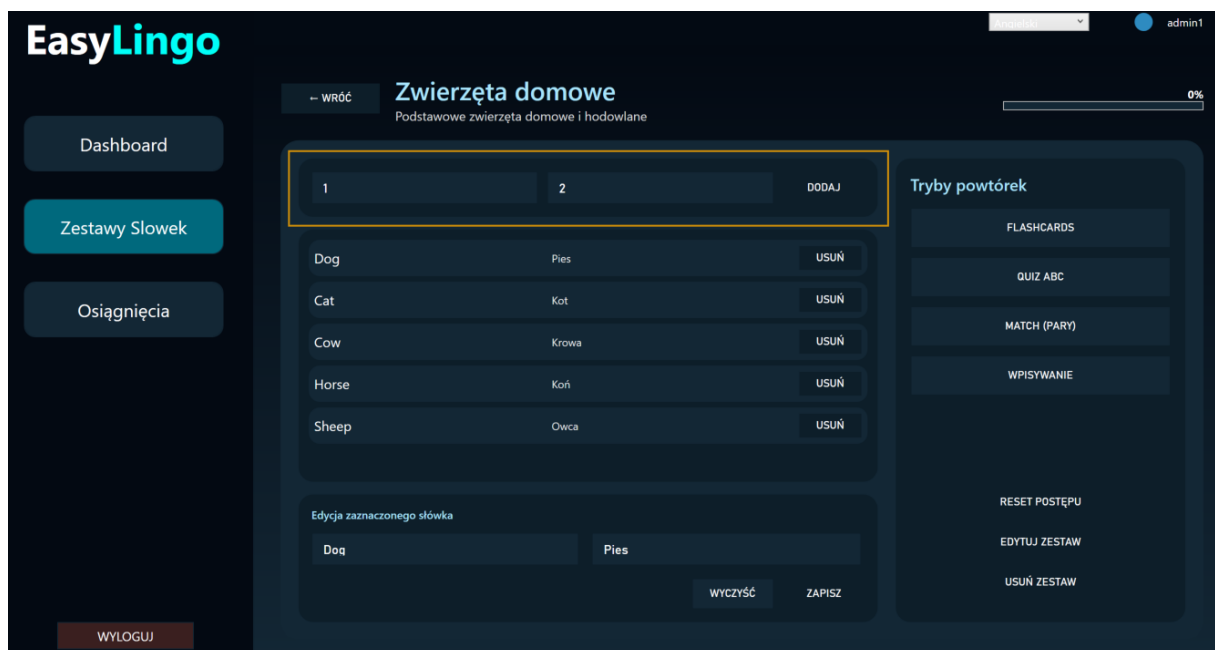
Opis

Kategoria

ANULUJ DODAJ

3. Adding a Term

Left-clicking a set in the *Zestawy Słówek* tab opens the set details view with a list of terms. To add a term, the user enters the term (1) and definition (2) in the input fields:



EasyLingo

admin1

0%

← WRÓĆ **Zwierzęta domowe**
Podstawowe zwierzęta domowe i hodowlane

1 2 DODAJ

Dog	Pies	USUŃ
Cat	Kot	USUŃ
Cow	Krowa	USUŃ
Horse	Koń	USUŃ
Sheep	Owca	USUŃ

Edycja zaznaczonego słowa

Dog Pies WYCZYŚĆ ZAPISZ

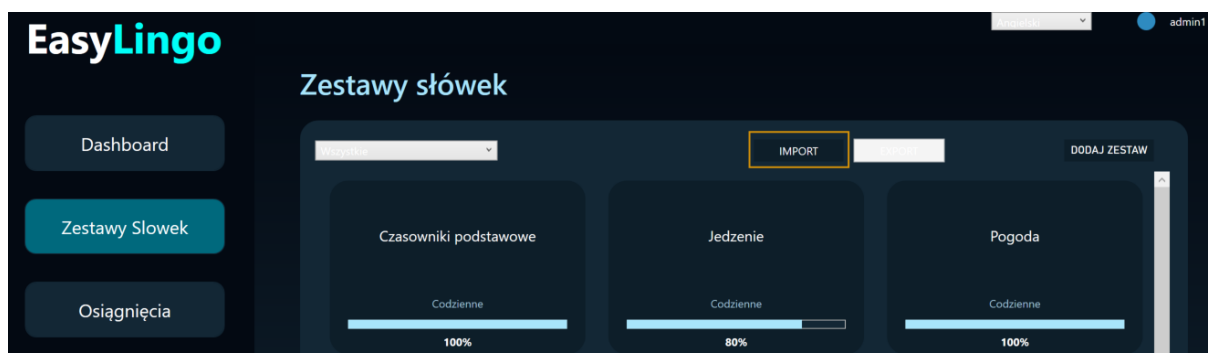
Tryby powtórek

- FLASHCARDS
- QUIZ ABC
- MATCH (PARY)
- WPISYWANIE
- RESET POSTĘPU
- EDYTUJ ZESTAW
- USUŃ ZESTAW

WYLOGUJ

4. Importing a Set

A JSON file must be created according to the structure described in the [JSON File Structure](#) section. Then, the IMPORT button should be clicked and the file selected.



5. Exporting a Set

The user must right-click a set and then click the EXPORT button.

